MIL-E-81099A (AS)
21 July 1966
Superseding
MIL-E-81099(WEP)
13 July 1964

MILITARY SPECIFICATION

ELECTRIC DEVICES, SIMPLE,
GENERAL SPECIFICATION FOR

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Items. - This specification gives general requirements for electric devices for which the detail requirements are given by standards only, as the detail requirements for the devices are not complex enough to warrant issuance of detail specifications for them.

1.2 Classification. - The devices are identified by the MS part numbers shown on the standard.

2. APPLICABLE DOCUMENTS

2.1 Issues. - The following documents, of the issues in effect on the date of the invitation for bids, are cited by this specification for certain requirements. Whenever any requirement of any document cited by this specification conflicts with any requirement explicitly stated in this specification, the requirement explicitly stated in this specification shall prevail.

MILITARY SPECIFICATIONS	
MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
FEDERAL STANDARD	
FED-STD-595	Colors
MILITARY STANDARD	
MIL-STD-130	Identification Marking of U.S. Military Property

2.2 Obtaining Copies. - When requesting copies of specifications and standards, give both number and title, and, when the documents are for use with a particular Invitation for Bids, the source, number, and date of the IFB. Copies of this specification and its applicable documents may be obtained from the Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Qualification. - The devices furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

FSC 6150

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3.2 Dimensions. - Each device shall conform to the dimensions shown on the standard. Where only maximum dimensions are shown on the standard, the device need not have the shape shown on the standard, but the device, including all protrusions such as rivet- and screw-heads, shall be contained within the outline shown on the standard.

3.3 Special Requirements. - Each device shall conform to any special requirements called out on the standard.

3.4 Design Details. - Where the standard shows design details but does not specifically require them, they are shown as typical and for information only - such design details are optional.

3.5 Parts. - Each device shall include the parts called out on the standard.

3.6 Color. - The color of all parts of the device which are visible when the device is installed in accordance with the standard shall conform to FED-STD-595 for Color Number 17875.

3.7 Weight. - The weight of each device shall not exceed the value shown on the standard.

3.8 Identification Marking. - Parts, subassemblies, and devices shall be marked for identification in accordance with MIL-STD-130.

3.9 Suitability for Use. - The devices shall be capable of passing the tests of Section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection or testing called out in the specification when it considers such inspection or testing necessary to assure that the devices being furnished under the contract conform to this specification and the standard.

4.2 Qualification Testing. - A sample of devices to be tested for qualification to be offered on a bid shall consist of at least three devices of a given manufacturer's part number. The devices shall be subjected to all of the tests of this specification and the standard.

4.3 Acceptance Testing. - Devices offered for acceptance under a contract shall be tested as follows:

4.3.1 Individual Tests. - Each device shall be subjected to the individual tests called out on the standard.

4.3.2 Sampling Tests. - Devices selected at random by the government inspector, or automatically, in accordance with the sampling plan called out on the standard, shall be subjected to the sampling tests called out on the standard.

4.3.3 Sampling Test Failure. - When a device fails a sampling test, the manufacturer may perform that test as an individual test until the inspector directs that it revert to sampling test status, or suspend production until the failure is corrected to the satisfaction of the inspector.

4.4 Rejection and Resubmission. - When a device fails a test, it may be reworked and resubmitted, or a new device may be submitted in place of it. A device which has been reworked and resubmitted may, at the discretion of the inspector, be subjected to any applicable tests of this specification beside

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those which it had not passed before it was reworked.

4.5 Test Conditions. - Unless otherwise specified in the description of the test, all tests of this specification shall be performed in the conditions called out on the standard and the following:

4.5.1 Attitude. - The device shall be in the attitude in which it is shown on the standard.

4.5.2 Temperature. - The ambient temperature and the temperature of the device at the beginning of the test shall be 80±15°F. This is not considered a controlled temperature.

4.5.3 Evaluation of Equipment Test. - The Evaluation test shall be performed as the Evaluation of Equipment Test of MIL-E-5272.

4.5.4 Controlled Temperature Tests. - When a temperature other than 80±15°F is specified, the device shall be kept in a chamber maintained at the specified temperature, ±5°F, for 24±0.6 hours, and then, while remaining in the chamber maintained at this temperature, shall be subjected to the test.

4.5.5 Order of Tests. - The tests on each device shall be performed in numerical order.

4.6 Tests. -

4.6.1 Dimensions. - The device shall be measured for conformance to 3.2. Measurements performed during manufacture are acceptable in fulfillment of this requirement.

4.6.2 Examination. - The device shall be examined, tested, etc., as necessary to determine conformance to all requirements of the standard for which there are no specific tests.

4.6.3 Weight. - The device shall be weighed for conformance to 3.7.

4.6.4 Evaluation . - The device shall be subjected to the Evaluation test called out on the standard.

4.6.5 Evaluation at -65°F. - The device shall be subjected to an Evaluation test at -65°F.

4.6.6 Evaluation at 140°F. - The device shall be subjected to an Evaluation test at 140°F.

4.6.7 Temperature Shock. - If the device is not made entirely of metal, it shall be subjected to the temperature shock test procedure I of MIL-E-5272.

4.6.8 Humidity. - The device shall be subjected to the Humidity Test Procedure III of MIL-E-5272.

4.6.9 Fungus. - If the device is not made entirely of metal, it shall be subjected to the Fungus Test Procedure I of MIL-E-5272.

4.6.10 Salt Spray. - The device shall be subjected to synthetic sea water spray in accordance with the Salt Spray Test Procedure I of MIL-E-5272 modified as follows: The device shall be placed in the salt spray chamber for 4±0.1 hours. It shall then be removed from the chamber and kept out of the chamber for 30±1 minutes. It shall then be reinstalled in the chamber and subjected to 46±0.5 additional hours of salt spray. It shall then be removed from the chamber and washed with a gentle stream of water whose temperature shall not exceed 100°F. The device shall be dried for 24±1 hours in an ambient temperature of 125±10°F and shall then be subjected to performance test. There shall be no evidence of excessive corrosion. The synthetic sea water shall be prepared as follows:

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INGREDIENT	GRAMS PER LITER OF WATER
Sodium Chloride	24.540
Magnesium Chloride	11.110
Sodium Sulphate	4.094
Calcium Chloride	1.159
Potassium Chloride	0.695
Sodium Bicarbonate	0.201
Potassium Bromide	0.101
Boric Acid	0.027
Strontium Chloride	0.042
Sodium Fluoride	0.003

The pH of the solution shall be adjusted, if necessary, to 8.2 with 0.1 normal sodium carbonate solution.

5. PREPARATION FOR DELIVERY

5.1 Level. - The devices shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555. For Level A packaging, the method of packaging shall be Method 1C of MIL-P-116.

6. NOTES

6.1 Provisions for Qualification. - Awards will be made only for products which have been approved for inclusion in the applicable Qualified Products List, whether or not the products have been placed on the list. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have products which they intend to offer on bids tested for qualification, so that they will be eligible to receive contracts and orders for the devices covered by this specification. The Naval Air Systems Command, Navy Department, Washington 25, D. C., is responsible for the Qualified Products List applicable to this specification, and information about qualification of products under this specification may be obtained by writing to that activity, Attention AIR-5368.

6.2 Ordering Data. - Procurement documents should call out the number, date, and title of this specification; the amendment number and date of the latest amendment to this specification, if any; the number (including the revision letter, if any), date, and title of the standard (all of the latest revision, if any); the MS part number of the device; and the levels of preservation, packaging, packing, and marking.

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